

## Lecture 35. GMRES

In the last lecture we showed how the Arnoldi process can be used to find eigenvalues. Here we show that it can also be used to solve systems of equations  $Ax = b$ . The standard algorithm of this kind is known as GMRES, which stands for “generalized minimal residuals.”

### Residual Minimization in $\mathcal{K}_n$

As in the last two lectures, let  $A \in \mathbb{C}^{m \times m}$  be a square matrix and  $b \in \mathbb{C}^m$  a vector, and let  $\mathcal{K}_n$  denote the Krylov subspace  $\langle b, Ab, \dots, A^{n-1}b \rangle$  of (33.5). Now, however, we assume that  $A$  is nonsingular, for our goal is to solve a system of equations  $Ax = b$ . It will be convenient to have a notation for the exact solution of this problem:  $x_* = A^{-1}b$ .

The idea of GMRES is a one-liner. At step  $n$ , we shall approximate  $x_*$  by the vector  $x_n \in \mathcal{K}_n$  that minimizes the norm of the residual  $r_n = b - Ax_n$ . In other words, we shall determine  $x_n$  by solving a least squares problem, illustrated in Figure 35.1.

The obvious way to solve this least squares problem would be as follows.

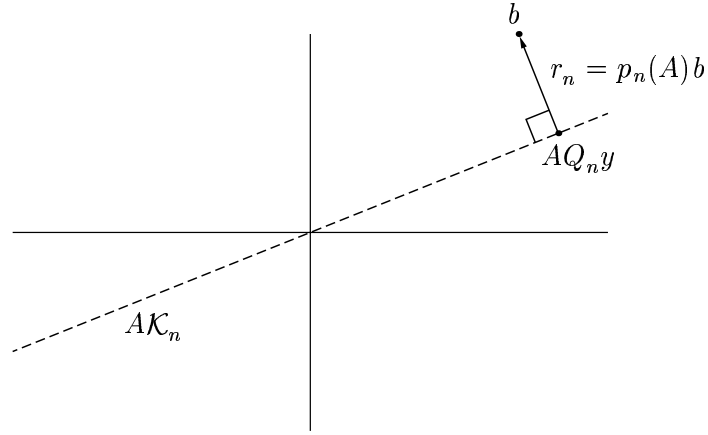


Figure 35.1. *The least squares polynomial approximation problem underlying GMRES: minimize the residual norm  $\|r_n\|$ . Compare Figure 34.1.*

Let  $K_n$  be the  $m \times n$  Krylov matrix (33.6), so that we have

$$AK_n = \left[ \begin{array}{c|c|c|c} Ab & A^2b & \cdots & A^n b \end{array} \right]. \quad (35.1)$$

The column space of this matrix is  $AK_n$ . Thus our problem is to find a vector  $c \in \mathbb{C}^n$  such that

$$\|AK_n c - b\| = \text{minimum}, \quad (35.2)$$

where  $\|\cdot\| = \|\cdot\|_2$ , as always in this part of the book. This could be done by means of a QR factorization of  $AK_n$ , in analogy to (33.7). Once  $c$  is found, we would set  $x_n = K_n c$ .

The procedure just described is numerically unstable, however, and it constructs a factor  $R$  that is not needed. Here is what is actually done instead. We use the Arnoldi iteration (Algorithm 33.1) to construct a sequence of Krylov matrices  $Q_n$  whose columns  $q_1, q_2, \dots$  span the successive Krylov subspaces  $\mathcal{K}_n$ . Thus we write  $x_n = Q_n y$  instead of  $x_n = K_n c$ . Instead of (35.2), our least squares problem is to find a vector  $y \in \mathbb{C}^n$  such that

$$\|AQ_n y - b\| = \text{minimum}. \quad (35.3)$$

Superficially, the problem (35.3) has dimensions  $m \times n$ . In actuality, however, because of the special structure of Krylov subspaces, it is essentially of dimensions  $(n+1) \times n$ . We can reveal this as follows. Applying (33.3) transforms the equation to

$$\|Q_{n+1} \tilde{H}_n y - b\| = \text{minimum}. \quad (35.4)$$

Now both vectors inside the norm are in the column space of  $Q_{n+1}$ . Therefore, multiplying on the left by  $Q_{n+1}^*$  does not change that norm. Thus a further equivalent problem is

$$\|\tilde{H}_n y - Q_{n+1}^* b\| = \text{minimum}. \quad (35.5)$$

Finally, we note that by construction of the Krylov matrices  $\{Q_n\}$ ,  $Q_{n+1}^* b$  is equal to  $\|b\|e_1$ , where  $e_1 = (1, 0, 0, \dots)^*$  as usual. Thus we reach at last the final form of the GMRES least squares problem:

$$\|\tilde{H}_n y - \|b\|e_1\| = \text{minimum}. \quad (35.6)$$

At step  $n$  of GMRES we solve this problem for  $y$ , then set  $x_n = Q_n y$ .

## Mechanics of GMRES

This completes our derivation of the GMRES algorithm. A high-level description is as follows.

### Algorithm 35.1. GMRES

$$q_1 = b/\|b\|$$

**for**  $n = 1, 2, 3, \dots$

*< step  $n$  of Arnoldi iteration, Algorithm 33.1 >*

Find  $y$  to minimize  $\|\tilde{H}_n y - \|b\|e_1\|$  ( $= \|r_n\|$ )

$$x_n = Q_n y.$$

At each step, GMRES minimizes the norm of the residual  $r_n = b - Ax_n$  over all vectors  $x_n \in \mathcal{K}_n$ . The quantity  $\|r_n\|$  is computed in the course of finding  $y$ ; one need not calculate it explicitly from  $x_n$ .

The inner, “Find  $y$ ” step of Algorithm 35.1 is an  $(n+1) \times n$  matrix least squares problem with Hessenberg structure. It can be solved via QR factorization in the usual manner described in Lecture 11, at a cost of  $O(n^2)$  flops, thanks to the Hessenberg structure. In addition, it is possible to save further work by a more specialized approach. Rather than construct QR factorizations of the successive matrices  $\tilde{H}_1, \tilde{H}_2, \dots$  independently, one can use an updating process to get the QR factorization of  $\tilde{H}_n$  from that of  $\tilde{H}_{n-1}$ . All that is required is a single Givens rotation (Exercise 10.4) and  $O(n)$  work. See Exercise 35.4.

## GMRES and Polynomial Approximation

In the last lecture, we showed that the calculation of eigenvalues by the Arnoldi iteration is related to the approximation problem (34.3): find  $p^n \in P^n$  to minimize  $\|p^n(A)b\|$ , where  $P^n$  denotes the set of monic polynomials of degree  $n$ .

The GMRES iteration also solves an approximation problem, the only difference being that the space of polynomials is now

$$P_n = \{\text{polynomials } p \text{ of degree } \leq n \text{ with } p(0) = 1\}. \quad (35.7)$$

Expressed in terms of polynomial coefficients, our normalization is now  $c_0 = 1$  rather than  $c_n = 1$ . Notationally, we have changed from a superscript to a subscript as a reminder of the change from normalization at  $z = \infty$  to normalization at  $z = 0$ .

Here is how GMRES can be reduced to polynomial approximation in  $P_n$ . The iterate  $x_n$  can be written

$$x_n = q_n(A)b, \quad (35.8)$$

where  $q$  is a polynomial of degree  $n - 1$ ; its coefficients are the entries of the vector  $c$  of (35.2). The corresponding residual  $r_n = b - Ax_n$  is  $r_n = (I - Aq_n(A))b$ , where  $p_n$  is the polynomial defined by  $p_n(z) = 1 - zq(z)$ . In other words, we have

$$r_n = p_n(A)b \quad (35.9)$$

for some polynomial  $p_n \in P_n$ . The GMRES process chooses the coefficients of  $p_n$  to minimize the norm of this residual. Thus we have shown that GMRES solves the following approximation problem successively for  $n = 1, 2, 3, \dots$

**GMRES Approximation Problem.** Find  $p_n \in P_n$  such that

$$\|p_n(A)b\| = \text{minimum}. \quad (35.10)$$

Like the Arnoldi iteration, GMRES satisfies certain invariance properties. The following theorem is easily proved from (35.10).

**Theorem 35.1.** *Let the GMRES iteration be applied to a matrix  $A \in \mathbb{C}^{m \times m}$  as described above.*

*Scale-invariance. If  $A$  is changed to  $\sigma A$  for some  $\sigma \in \mathbb{C}$ , and  $b$  is changed to  $\sigma b$ , the residuals  $\{r_n\}$  change to  $\{\sigma r_n\}$ .*

*Invariance under unitary similarity transformations. If  $A$  is changed to  $UAU^*$  for some unitary matrix  $U$ , and  $b$  is changed to  $Ub$ , the residuals  $\{r_n\}$  change to  $\{U^*r_n\}$ .*

GMRES is not invariant under translation, since the normalization  $p(0) = 1$  involves the translation-dependent point 0. On the contrary, its behavior depends strongly on the position of the origin—loosely speaking, on the condition number of  $A$ .

## Convergence of GMRES

How quickly does GMRES converge? How many steps  $n$  must be taken before  $\|r_n\|/\|b\|$  is reduced to a satisfactory level such as  $10^{-3}$  or  $10^{-16}$ ? As a practical matter, this often becomes the question of designing a good preconditioner (Lecture 40). Mathematically, the problem is to investigate what properties of  $A$  determine the size of  $\|r_n\|$ .

We begin with two observations. The first is that GMRES converges monotonically:

$$\|r_{n+1}\| \leq \|r_n\|. \quad (35.11)$$

The reason is that  $\|r_n\|$  is as small as possible for the subspace  $\mathcal{K}_n$ ; by enlarging  $\mathcal{K}_n$  to the space  $\mathcal{K}_{n+1}$ , we can only decrease the residual norm, or at worst leave it unchanged. (Note that  $P_n \subseteq P_{n+1}$ , whereas  $P^n \not\subseteq P^{n+1}$ .) The second is that after at most  $m$  steps the process must converge, at least in the absence of rounding errors:

$$\|r_m\| = 0. \quad (35.12)$$

For generic data  $A$  and  $b$ , this will happen because  $\mathcal{K}_m = \mathbb{C}^m$ , and in special cases, if  $x_*$  happens to lie in  $\mathcal{K}_n$  for some  $n < m$ , convergence will occur earlier. Equation (35.12) is useful as a reminder of the mathematics of GMRES, but it has little practical content, since a GMRES iteration must converge to satisfactory precision in  $n \ll m$  steps if it is to be useful.

To obtain more useful information about convergence, we must turn to the polynomial approximation problem (35.10). We know that  $\|r_n\| = \|p_n(A)b\| \leq \|p_n(A)\| \|b\|$  is minimal. Except for problems where the right-hand side  $b$  has special structure related to that of  $A$ , the critical factor that determines the size of this quantity is usually  $\|p_n(A)\|$ . That is, what determines the convergence rate of GMRES is generally the inequality

$$\frac{\|r_n\|}{\|b\|} \leq \inf_{p_n \in P_n} \|p_n(A)\|. \quad (35.13)$$

This brings us to the mathematically elegant question: given a matrix  $A$  and a number  $n$ , how small can  $\|p_n(A)\|$  be? This question is the basis of almost all analyses of convergence of Krylov subspace iterations for solving systems of equations.

## Polynomials Small on the Spectrum

Given  $A$  and  $n$ , how small can  $\|p_n(A)\|$  be? The standard way of obtaining estimates is to look for polynomials  $p(z)$  that are as small as possible on the spectrum  $\Lambda(A)$ , while still satisfying  $p(0) = 1$ . If  $p$  is a polynomial and  $S$  is a set in the complex plane, let us define the scalar  $\|p\|_S$  by

$$\|p\|_S = \sup_{z \in S} |p(z)|. \quad (35.14)$$

Suppose  $A$  is diagonalizable, satisfying  $A = V\Lambda V^{-1}$  for some nonsingular matrix  $V$  and diagonal matrix  $\Lambda$ . Then we have

$$\|p(A)\| \leq \|V\| \|p(\Lambda)\| \|V^{-1}\| = \kappa(V) \|p\|_{\Lambda(A)}. \quad (35.15)$$

Combining this result with (35.13) gives the following basic theorem on convergence of GMRES.

**Theorem 35.2.** *At step  $n$  of the GMRES iteration, the residual  $r_n$  satisfies*

$$\frac{\|r_n\|}{\|b\|} \leq \inf_{p_n \in P_n} \|p_n(A)\| \leq \kappa(V) \inf_{p_n \in P_n} \|p_n\|_{\Lambda(A)}, \quad (35.16)$$

where  $\Lambda(A)$  is the set of eigenvalues of  $A$ ,  $V$  is a nonsingular matrix of eigenvectors (assuming  $A$  is diagonalizable), and  $\|p_n\|_{\Lambda(A)}$  is defined by (35.14).

This theorem can be summarized in words as follows. If  $A$  is not too far from normal in the sense that  $\kappa(V)$  is not too large, and if properly normalized degree  $n$  polynomials can be found whose size on the spectrum  $\Lambda(A)$  decreases quickly with  $n$ , then GMRES converges quickly.

**Example 35.1.** Here is a numerical example. Let  $A$  be a  $200 \times 200$  matrix whose entries are independent samples from the real normal distribution of mean 2 and standard deviation  $0.5/\sqrt{200}$ . In MATLAB,

$$m = 200; A = 2*\text{eye}(m) + 0.5*\text{randn}(m)/\text{sqrt}(m). \quad (35.17)$$

Figure 35.2 shows the eigenvalues of  $A$ , a set of points roughly uniformly distributed in the disk of radius  $1/2$  centered at  $z = 2$  (Exercise 12.3). Figure 35.3 shows the convergence curve for the GMRES iteration applied to the problem  $Ax = b$ , where  $b = (1, 1, \dots, 1)^*$ . The convergence in this case is extraordinarily steady at a rate approximately  $4^{-n}$ . The reason for this is not hard to spot. Since the spectrum of  $A$  approximately fills the disk indicated,  $\|p(A)\|$  is approximately minimized by the choice  $p(z) = (1 - z/2)^n$ . Since  $I - A/2$  is a random matrix scaled so that its spectrum approximately fills the disk of radius  $1/4$  about 0, we have  $\|p(A)\| = \|(I - A/2)^n\| \approx 4^{-n}$ . This matrix  $A$  is well-conditioned, with condition number  $\kappa(A) \approx 2.03$ . The deviation from normality is modest, with  $\kappa(V) \approx 141$ .

Figure 35.3 illustrates the convergence of matrix iterations under favorable circumstances—when the matrix  $A$  is well behaved (which often means a good preconditioner has been applied). We see that six-digit accuracy is achieved after ten iterations, at a cost of approximately  $10 \times 2m^2 = 8.0 \times 10^5$  flops, since the work is dominated by the matrix-vector multiplication at each step. Solving the same system by Gaussian elimination would require  $\frac{2}{3}m^3 \approx 5.3 \times 10^6$  flops. This improvement by a factor close to 7 was achieved even though  $A$  has no sparsity to take advantage of and even though the dimension  $m = 200$  is

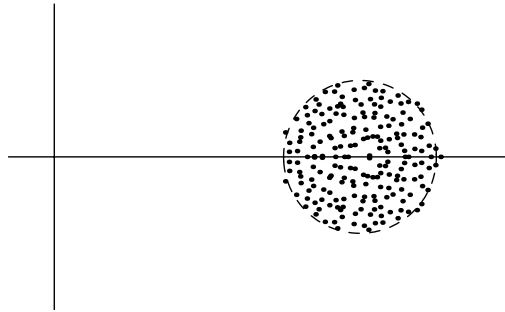


Figure 35.2. *Eigenvalues of the  $200 \times 200$  matrix  $A$  of (35.17). The dashed curve is the circle of radius  $1/2$  with center  $z = 2$  in  $\mathbb{C}$ . The eigenvalues are approximately uniformly distributed within this disk.*

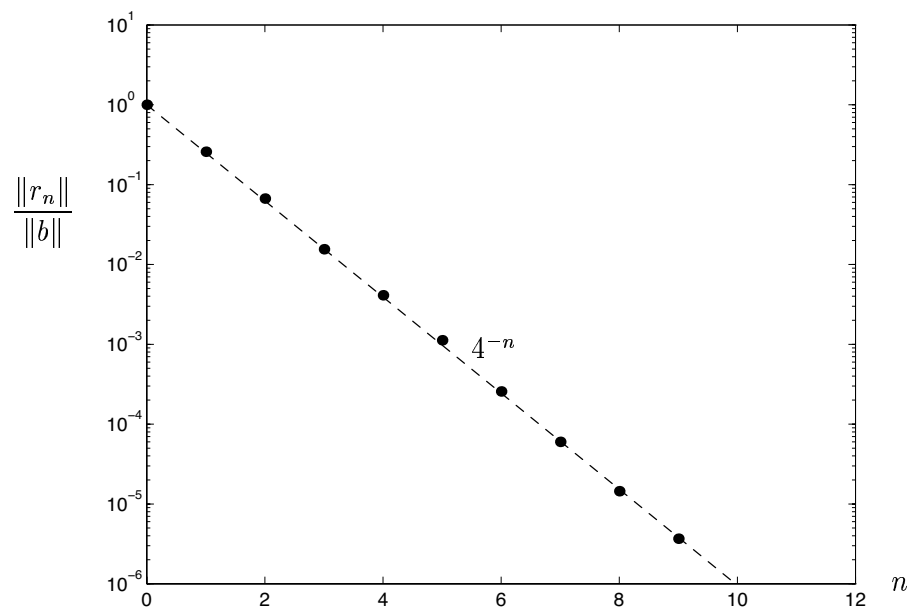


Figure 35.3. *GMRES convergence curve for the same matrix  $A$ . This rapid, steady convergence is illustrative of Krylov subspace iterations under ideal circumstances, when  $A$  is a well-behaved (or well-preconditioned) matrix.*

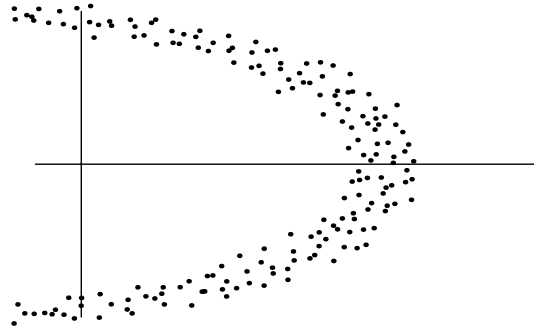


Figure 35.4. *Eigenvalues of a  $200 \times 200$  matrix, like that of (35.17) except with a modified diagonal. Now the eigenvalues surround the origin on one side.*

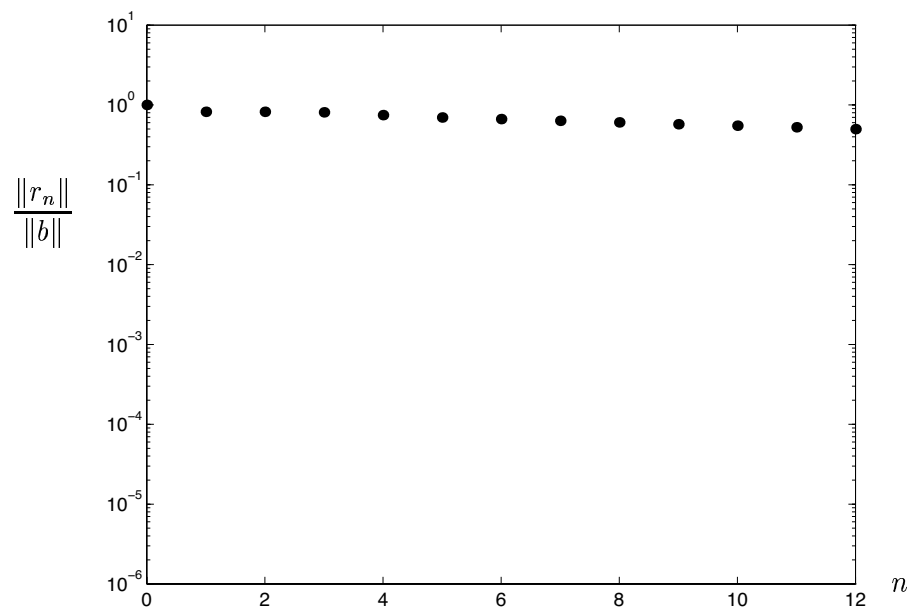


Figure 35.5. *GMRES convergence curve for the matrix of Figure 35.4. The convergence has slowed down greatly. When an iterative method stagnates like this, it is time to look for a better preconditioner.*

not high. For the same example with  $m = 2000$ , GMRES would beat Gaussian elimination by a factor more like 70. For a  $2000 \times 2000$  matrix with similar spectral properties but 90% or 99% sparsity, the factor would improve to on the order of 700 or 7000, respectively, and the storage required by GMRES would also diminish dramatically.  $\square$

**Example 35.2.** If the eigenvalues of a matrix “surround the origin,” on the other hand, such rapid convergence cannot be expected. Figures 35.4–35.5 present an example. The matrix is now  $A' = A + D$ , where  $A$  is the matrix of (35.17) and  $D$  is the diagonal matrix with complex entries

$$d_k = (-2 + 2 \sin \theta_k) + i \cos \theta_k, \quad \theta_k = \frac{k\pi}{m-1}, \quad 0 \leq k \leq m-1.$$

As is evident in Figure 35.4, the eigenvalues now lie in a semicircular cloud that bends around the origin. The convergence rate is much worse than before, making the iterative computation no better than Gaussian elimination for this problem. The condition numbers are now  $\kappa(A) \approx 4.32$  and  $\kappa(V) \approx 54.0$ , so the deterioration in convergence cannot be explained by conditioning alone; it is the locations of the eigenvalues, not their magnitudes (or those of the singular values) that are causing the trouble. If the arc extended much further around the spectrum, the convergence would worsen further.  $\square$

## Exercises

**35.1.** Show that if  $S \subseteq \mathbb{C}$  contains infinitely many points, then (35.14) defines a norm on the vector space of all polynomials with complex coefficients. Explain what goes wrong if  $S$  has only finitely many points.

**35.2.** (a) Let  $S \subseteq \mathbb{C}$  be a set whose convex hull contains 0 in its interior. That is,  $S$  is contained in no half-plane disjoint from the origin. Show that there is no  $p \in P_1$  (i.e., no polynomial  $p$  of degree 1 with  $p(0) = 1$ ) such that  $\|p\|_S < 1$ .

(b) Let  $A$  be a matrix, not necessarily normal, whose spectrum  $\Lambda(A)$  has the property (a). Show that there is no  $p \in P_1$  such that  $\|p(A)\| < 1$ .

(c) Though the convergence in Figure 35.5 is slow, it is clear that  $\|r_1\| < \|r_0\|$ . Explain why this does not contradict the result of (b). Describe what kind of polynomial  $p_1 \in P_1$  GMRES has probably found to achieve  $\|r_1\| < \|r_0\|$ .

**35.3.** The recurrence  $x_{n+1} = x_n + \alpha r_n = x_n + \alpha(b - Ax_n)$ , where  $\alpha$  is a scalar constant, is known as a *Richardson iteration*.

(a) What polynomial  $p(A)$  at step  $n$  does this correspond to?

(b) What choice of  $\alpha$  would you recommend for the matrix  $A$  of Figure 35.2, and what would you expect to be the corresponding convergence rate?

(c) Same questions for the matrix of Figure 35.4.

**35.4.** (a) Describe an  $O(n^2)$  algorithm based on QR factorization by Givens rotations (Exercise 10.4) for solving the least squares problem of Algorithm 35.1.

(b) Show how the operation count can be improved to  $O(n)$ , as mentioned on p. 268, if the problem for step  $n - 1$  has already been solved.

**35.5.** Our statement of the GMRES algorithm (Algorithm 35.1) begins with the initial guess  $x_0 = 0$ ,  $r_0 = b$ . (The same applies to CG and BCG, Algorithms 38.1 and 39.1.) Show that if one wishes to start with an arbitrary initial guess  $x_0$ , this can be accomplished by an easy modification of the right-hand side  $b$ .

**35.6.** For larger values of  $n$ , the cost of GMRES in operations and storage may be prohibitive. In such circumstances a method called *k-step restarted GMRES* or *GMRES(k)* is often employed, in which, after  $k$  steps, the GMRES iteration is started anew with the current vector  $x_k$  as an initial guess.

(a) Compare the asymptotic operation counts and storage requirements of GMRES and GMRES( $k$ ), for fixed  $k$  and increasing  $n$ .

(b) Describe an example in which GMRES( $k$ ) can be expected to converge in nearly as few iterations as GMRES (hence much faster in operation count).

(c) Describe another example in which GMRES( $k$ ) can be expected to fail to converge, whereas GMRES succeeds.